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# MyoPong: Goal-Conditioned Synergistic Action and Latent Exploration for Musculoskeletal Table Tennis Control with Training-Free High-Level Policy

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## Abstract

Controlling bio-inspired musculoskeletal systems (MSK) remains a significant challenge in robotics due to their high dimensionality, overactuated nature, and the sparse rewards typical of complex tasks like table tennis. We introduce MyoPong, a framework designed to tackle the MyoChallenge Table-Tennis task using a musculoskeletal model (myoArm and myoTorso) with up to 210 muscles. To overcome the challenges of sparse rewards and high dimensionality, the system combines a training-free, physics-based high-level planner with a low-level PPO actor integrated with muscle synergy extraction and latent exploration (the Lattice method). Experimental results show that while standard PPO baselines fail completely, the proposed hierarchical architecture achieves a near-optimal 94% success rate. The findings demonstrate that combining hierarchical structures with inductive biases in the action and exploration spaces is essential for mastering musculoskeletal control, with the Lattice-based approach emerging as the most energy-efficient solution for balancing task success and minimal control effort. Project details and demonstrations on our official website: <https://myopong.github.io><sup>1</sup>.

## 1 Introduction

Developing motoric intelligence, the capacity to perceive, reason, and execute precise physical actions in dynamic environments, remains a fundamental milestone for autonomous systems. While recent breakthroughs in perception [1, 2] and large-scale reasoning [3, 4] have transformed the landscape of artificial intelligence, a profound gap remains in physical dexterity. Modern robotic systems often excel at repetitive, high-precision tasks but lack the adaptability and compliance characteristic of biological control. In contrast, the human musculoskeletal (MSK) system seamlessly integrates sensory feedback with complex internal dynamics to perform agile movements [5]. Understanding and replicating this biological advantage requires addressing the inherent complexity of MSK motor control.

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<sup>1</sup>  Our code repository is available on <https://github.com/alfonsusrr/MyoPong>

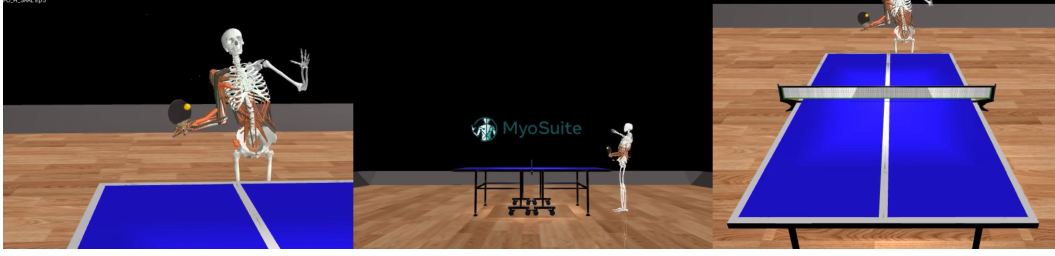


Figure 1: The 2025 MyoChallenge Table Tennis environment. The agent consists of an integrated *myoTorso* and *myoArm* (273 muscles total) tasked with intercepting randomly launched ping-pong balls and returning them across the net to a target zone.

MSK systems differ fundamentally from engineered motor systems in their structure and control principles. They comprise a redundant network of muscles and tendons acting on compliant joints, creating a high-dimensional control space. Biologically, this involves roughly 600 muscles acting on 300 joints [6], a configuration known as an overactuated system where multiple actuators contribute to a single degree of freedom (DoF) and individual muscles often span multiple joints. This redundancy, often referred to as Bernstein’s problem, creates an enormous action space that poses significant challenges, as countless activation patterns can result in the same physical trajectory [7].

To bridge the gap between biological and artificial control, the MyoSuite framework provides a physiologically realistic and computationally efficient platform for musculoskeletal simulation [8]. By modeling muscles and tendons with high fidelity, MyoSuite enables the study of contact-rich interactions beyond simple joint torques. The 2025 MyoChallenge Table-Tennis task serves as a rigorous benchmark for these challenges, requiring the coordination of a *myoArm* (63 muscles) and a *myoTorso* (210 muscles) integrated into a single agent. The objective is to return randomly launched ping-pong balls across a net to a target zone on the opponent’s side. This task is characterized by high-dimensional observations and exceptionally sparse rewards.

Prior attempts to solve such tasks using standard model-free deep reinforcement learning (DRL) often result in suboptimal "shortcut" behaviors. In these cases, agents may learn to make contact with the ball but fail to regulate the trajectory or direction, eventually leading to policy divergence as the task complexity increases<sup>2</sup>. These failures highlight the difficulty of exploring high-dimensional action spaces where unstructured noise often cancels out across redundant actuators.

In this work, we present *MyoPong*, a framework designed to overcome the dimensionality and reward sparsity of musculoskeletal table tennis. Our primary contributions are as follows:

- We propose a hierarchical control architecture that decouples strategic decision-making (High-Level Planner) from biomechanical execution (Low-Level Actor).
- We transform the complex, sparse-reward trajectory optimization problem into a dense-reward goal-reaching task with training-free physics-based policy, significantly improving training stability.
- We integrate an inductive bias into the exploration and action spaces, inspired by two previous works: Lattice and SAR, to promote coordinated muscle synergies rather than independent actuator noise.
- We evaluate this approach on the 2025 MyoChallenge benchmark, demonstrating its ability to coordinate 273 muscles for precise, dynamic ball returns.

## 2 Related Works

**Model-Free Deep Reinforcement Learning** Model-free DRL is a standard approach for complex control, with PPO [10] widely used for scalability, and SAC [11] or TD3 [12] utilized for stability in continuous spaces. While recurrent architectures like LSTMs help mitigate partial observability

<sup>2</sup>Refer to our results showed in our project proposal [9]

[13], these methods frequently struggle with the high-dimensional, contact-rich dynamics of musculoskeletal models. Recent benchmarks, such as the MyoChallenge [14, 15], demonstrate that standard PPO-based policies often fail to achieve coordinated control in complex motor tasks, highlighting deficiencies in sample efficiency and exploration.

**Curriculum Learning** Sparse rewards in high-dimensional motor tasks often lead to suboptimal exploration and unstable convergence [10]. Curriculum learning addresses this by structuring training through a sequence of tasks with increasing difficulty [16]. For musculoskeletal control, the Static-to-Dynamic Stabilization (SDS) curriculum [15] has proven particularly effective; by first mastering stabilization of intermediate poses before learning dynamic transitions, agents achieve smoother motion and significantly higher task performance than via direct training.

**Muscle Synergy Learning** Biological motor systems simplify their inherent redundancy through muscle synergies—low-dimensional activation patterns that coordinate multiple muscles [17, 18]. In DRL, these synergies are typically extracted using dimensionality reduction, such as PCA or matrix factorization, to compress the action space [15, 19]. Recent frameworks like DynSyn [20] and SAR [21] demonstrate that leveraging these bio-inspired correlations improves learning efficiency by grouping functionally redundant actuators into manageable control units.

**Structured Exploration in Overactuated Systems** Standard exploration strategies, such as independent Gaussian noise, are often ineffective in overactuated systems because uncorrelated perturbations tend to cancel each other out [22]. Structured exploration addresses this by injecting correlated noise to uncover coordinated behaviors. For example, DEP-RL [23] establishes dependencies across actions to facilitate discovery of motor patterns, while LATTICE [24] perturbs latent activations to enhance both performance and energy efficiency in complex motor tasks.

### 3 Methodology

#### 3.1 Hierarchical Decomposition: Strategic vs. Motoric Layers

MyoPong addresses high dimensionality and sparse rewards in musculoskeletal table tennis by decoupling strategic planning from motor execution. The framework utilizes a training-free High-Level Planner (HLP) to analytically derive intercept poses, and a goal-conditioned Low-Level Actor to translate these targets into muscle activations. This hierarchy masks complex contact and ball physics, effectively converting a sparse trajectory optimization problem into a dense-reward goal-reaching task.

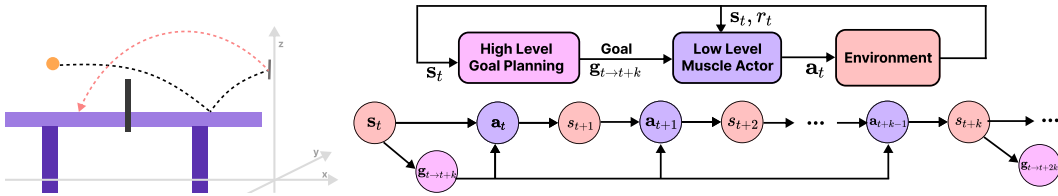


Figure 2: The Hierarchical Framework: (left) illustration of a ball trajectory and the derivation of the high-level planner policy; (right) the comprehensive goal-conditioned hierarchical architecture decoupling the high-level planner from the low-level muscle actor.

#### 3.2 High-Level Planner (HLP): Kinematic Trajectory Prediction

The HLP serves as a strategic heuristic module that analytically predicts the ball’s trajectory to determine the optimal intercept pose. Operating in the task-space using Newtonian kinematics, the HLP bypasses the need for high-dimensional exploration by calculating intercept requirements through a three-stage pipeline: longitudinal timing, vertical bounce dynamics, and 3D collision prediction.

**Longitudinal Impact Timing** The planner first determines the temporal horizon of the intercept by calculating the time required for the ball to reach the paddle’s operational  $x$ -plane. Assuming a

constant longitudinal velocity  $v_x$ , the impact time  $t_{imp}$  is derived as:

$$t_{imp} = \frac{p_x - b_x}{v_x} \quad (1)$$

where  $p_x$  represents the target  $x$ -coordinate of the strike zone and  $b_x$  is the current longitudinal position of the ball. This provides the temporal constraint required for the motor system to transition into an intercepting configuration.

**Table Interaction and Vertical Elasticity** To accurately predict the ball’s trajectory, the HLP accounts for the vertical velocity reversal during the table bounce. The post-bounce vertical velocity,  $v_{z,bounce}$ , is calculated by considering gravitational acceleration  $g$  and the coefficient of restitution  $e$  of the table:

$$v_{z,bounce} = -e(v_z - gt_{hit}) \quad (2)$$

where  $t_{hit}$  is the predicted time of table contact. This accounts for energy loss and ensures the predicted intercept height is physically consistent with table tennis dynamics.

**3D Collision Position Prediction** By integrating temporal and bounce constraints, the HLP predicts the 3D collision coordinates  $(x_{pred}, y_{pred}, z_{pred})$  at the intercept plane. The system of equations governing this prediction incorporates linear lateral motion and parabolic projectile motion:

$$\begin{cases} x_{pred} &= p_x \\ y_{pred} &= y_0 + v_y t_{imp} \\ z_{pred} &= z_{table} + v_{z,bounce} \cdot \Delta t - \frac{1}{2} g \Delta t^2 \end{cases}$$

where  $\Delta t = t_{imp} - t_{hit}$  represents the time elapsed between the table bounce and the predicted impact, and  $z_{table}$  denotes the height of the table.

**Target Pose and Goal Derivation** Once the collision position  $\mathbf{p}_{pred}$  is established, the HLP calculates the required paddle orientation (normal vector  $\mathbf{n}_p$ ) to ensure the ball is returned to a valid target zone. This is achieved by solving for the reflection vector:

$$\mathbf{n}_p = \frac{\mathbf{v}_{out} - \mathbf{v}_{in}}{\|\mathbf{v}_{out} - \mathbf{v}_{in}\|} \quad (3)$$

The HLP provides a 6-DoF goal  $\mathbf{g}_t = [x_{pred}, y_{pred}, z_{pred}, \mathbf{n}_p]$  to the LLA, updated every  $k$  simulation steps.

### 3.3 Low-Level Actor (LLA): Goal-Conditioned Formulation

The LLA is a goal-conditioned PPO agent designed to bridge the gap between the musculoskeletal state and the HLP’s target. The state space for the LLA is augmented with the current goal vector,  $s'_t = [s_t, \mathbf{g}_t]$ . We define a kinematic mapping  $\phi(s_t) \rightarrow \mathbb{R}^6$  that extracts the current 6-DoF pose of the end-effector from the state. The LLA is trained to maximize the expected discounted return:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[ \sum_{t=0}^T \gamma^t (R_{task}(s_t, a_t) - \lambda \|\phi(s_t) - \mathbf{g}_t\|^2) \right] \quad (4)$$

where  $R_{task}$  represents the sparse global reward for a successful rally and  $\lambda$  is a weighting coefficient for the dense goal-reaching penalty. This formulation ensures that the LLA focuses on the high-dimensional motor coordination required to achieve specific poses.

### 3.4 Task-Specific Muscle Synergy Extraction

To address the high dimensionality of the 273-muscle action space, we adapt the synergy extraction pipeline from the SAR framework [21]. We first collect a trajectory dataset of muscle activations  $\mathbf{M} \in [-1, 1]^{m \times T}$  during an exploratory "play phase" using a standard PPO policy. We then apply Principal Component Analysis (PCA) to capture the dominant variance in activations, followed by Independent Component Analysis (ICA) to isolate functionally independent synergy patterns,

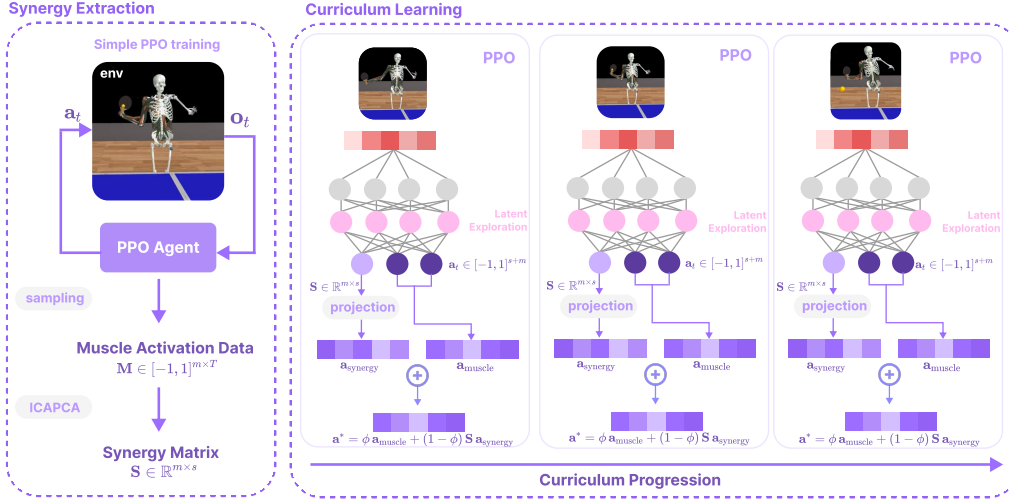


Figure 3: Overview of the full training pipeline. **Left:** The synergy extraction phase, where muscle activation data ( $\mathbf{M}$ ) is sampled from a baseline agent and processed via PCA and ICA to generate a task-specific synergy matrix ( $\mathbf{S}$ ). **Right:** The curriculum learning phase, illustrating the progression from simple to complex tasks. The policy architecture integrates **Latent Exploration** (pink layers) using the Lattice method to encourage coordinated noise, and a dual **action calculation** (purple blocks) that combines low-dimensional synergy activations with raw muscle commands. *Note: For experimental evaluation, latent exploration and synergy-based actions were implemented and tested as separate variants; a model combining both features was not evaluated.*

resulting in a synergy matrix  $\mathbf{S} \in \mathbb{R}^{m \times s}$ . Where  $s$  is determined based of the variance accounted for (VAF) all muscle activations trajectories.

The final action applied to the environment is computed as a linear combination of raw muscle activations and synergistic patterns:

$$\mathbf{a}^* = \phi \mathbf{a}_{\text{muscle}} + (1 - \phi) \mathbf{S} \mathbf{a}_{\text{synergy}} \quad (5)$$

where  $\mathbf{a}_{\text{synergy}}$  represents the low-dimensional latent control signal and  $\phi \in [0, 1]$  is a balancing coefficient. This adaptation allows the agent to rely on a lower-dimensional subspace for global movements while retaining the ability to perform fine-grained adjustments through individual muscle activations.

### 3.5 Structured Latent Exploration via the Lattice Method

In overactuated musculoskeletal systems, traditional exploration via independent Gaussian noise, where each of the 273 muscle channels is perturbed by uncorrelated  $\epsilon \sim \mathcal{N}(0, \sigma^2)$ , is often ineffective. Because multiple muscles frequently act as agonists or antagonists for the same joint, unstructured noise often leads to high-frequency tremors that are filtered by the low-pass dynamics of muscle-tendon units, resulting in negligible joint movement. To mitigate this, we adapt the Lattice exploration method [24], which injects structured, time-correlated noise into the neural network’s internal representations.

We define the network’s final latent layer as  $\mathbf{x} \in \mathbb{R}^d$  and the output weights as  $\mathbf{W} \in \mathbb{R}^{m \times d}$ . We introduce two learnable perturbation matrices: a muscle-space matrix  $\Sigma_a \in \mathbb{R}^{m \times d}$  and a latent-space matrix  $\Sigma_x \in \mathbb{R}^{d \times d}$ . The exploratory action  $\mathbf{a}$  is then calculated as:

$$\mathbf{a} = (\mathbf{W} + \mathbf{P}_a + \mathbf{W} \mathbf{P}_x) \mathbf{x} \quad (6)$$

where the perturbations are sampled from the learnable covariance structures such that  $(\mathbf{P}_a)_{i,j} \sim \mathcal{N}(0, (\Sigma_a)_{i,j})$  and  $(\mathbf{P}_x)_{i,j} \sim \mathcal{N}(0, (\Sigma_x)_{i,j})$ . These samples are held constant for a duration of  $T_{\text{sample}}$  timesteps. By adopting this structured approach, we ensure that perturbations remain consistent across both the action and feature spaces, facilitating the discovery of coordinated, multi-joint movements necessary for precise paddle-ball interactions.

### 3.6 Curriculum Learning Strategy

Learning complex musculoskeletal control policies from scratch is notoriously difficult due to the high dimensionality of the action space. To address this, we adopted a curriculum-based reinforcement learning strategy inspired by the winner of previous MyoChallenge [15], decomposing the training process into two distinct phases: Foundation Training and Advanced Adaptation.

**Phase 1: Foundation Training.** The primary objective of the first phase is to establish baseline competency across all model architectures. In this stage, the environment is simplified to reduce the variance of the state transition dynamics. In this phase, the target ball is initialized at random positions, but its launch velocity is held fixed across all episodes.

**Phase 2: Advanced Adaptation.** The second phase evaluates the robustness and generalization capability of the learned policies. This phase introduces significant domain randomization to simulate a stochastic, real-world physical environment. In this phase, while the initial ball position remains random, the environment dynamics are perturbed significantly. We introduce random variations in launch velocity, launch direction, table friction coefficients, and paddle mass.

## 4 Experiment Setup

To rigorously evaluate the proposed methods, we designed a comparative study involving six distinct model architectures. Our experiments focus on isolating the specific contributions of muscle synergy extraction and latent exploration (Lattice) within both simple and hierarchical reinforcement learning frameworks.

Table 1: Model Architectures and Feature Comparison

| Model Name              | Hierarchical | Synergy | Lattice |
|-------------------------|--------------|---------|---------|
| Standard PPO (Baseline) |              |         |         |
| PPO + Synergy           |              | ✓       |         |
| PPO + Lattice           |              |         | ✓       |
| Hierarchical PPO        | ✓            |         |         |
| Hierarchical + Synergy  | ✓            | ✓       |         |
| Hierarchical + Lattice  | ✓            |         | ✓       |

**Implementation Details** All agents were implemented using the Stable-Baselines3 (SB3) framework. To ensure stable learning dynamics, we applied observation and reward normalization across all experiments. Additionally, we utilized seeding to control randomness and ensure reproducibility.

The specific hyperparameter configurations used for the policy and value networks are detailed in Table 2. We employed a linear learning rate decay schedule to facilitate convergence.

Table 2: Hyperparameter settings and network architecture.

| Parameter                   | Value                                         |
|-----------------------------|-----------------------------------------------|
| Framework                   | Stable-Baselines3 (SB3)                       |
| Policy Network Architecture | $2 \times 512$ (Hidden Layers $\times$ Units) |
| Value Network Architecture  | $2 \times 256$ (Hidden Layers $\times$ Units) |
| Learning Rate               | $1 \times 10^{-4}$ (with linear decay)        |
| Initial Log Std             | -0.5                                          |
| Clip Range                  | 0.2                                           |

**Training Protocol** Experiments follow the curriculum defined in Section 3.6, structured into two phases: (1) **Foundation Benchmarking.** All six model architectures are trained in a fixed-velocity environment to establish performance baselines and evaluate the learning efficiency of the muscle

synergy and Lattice exploration modules. (2) **Dynamic Adaptation**. To test robustness, the top-performing **Hierarchical + Synergy** and **Hierarchical + Lattice** models are further trained in dynamic environments featuring variable velocity commands.

**Evaluation Metrics** We evaluate performance using **Success Rate (SR)**, representing the proportion of  $N$  episodes meeting the task criteria:

$$SR = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{success}_i) \quad (7)$$

where  $\mathbb{I}(\cdot)$  is the indicator function. Efficiency is measured by **Control Effort (CE)**, defined as the temporal average of the control signals  $\ell_2$  norm:

$$CE = \frac{1}{T} \sum_{t=1}^T \|\mathbf{a}_t\|_2 \quad (8)$$

Lower  $CE$  indicates more energy efficient motor control.

## 5 Experimental Results

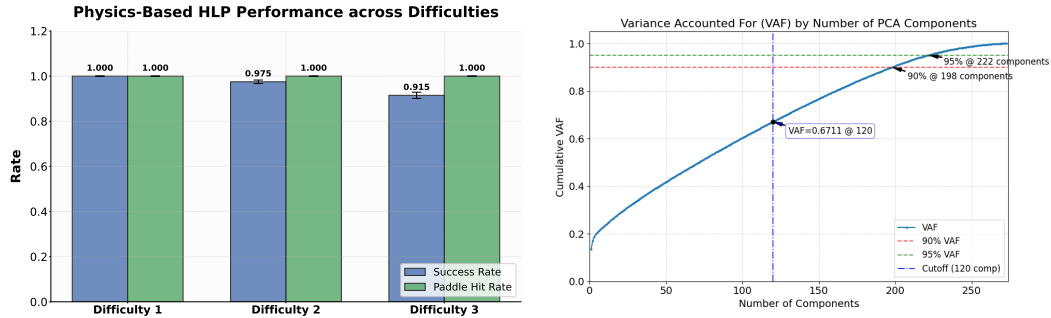


Figure 4: Foundational Components: (left) validation of the physics-based High-Level Policy (HLP) on simplified 6-DOF paddle control, achieving 91.5% success; (right) PCA analysis showing Cumulative Variance Accounted For (VAF) to determine the optimal dimensionality (120 components) for muscle synergy control.

### 5.1 Synergy Explainability and Dimensionality Reduction

The complexity of the musculoskeletal system necessitates a reduction in control dimensionality to make reinforcement learning tractable. As shown in the PCA analysis of muscle activation patterns (Figure 4), the system exhibits high redundancy. While 198 components are required to capture 90% of the variance, we established a cutoff at 120 components, which accounts for 67.11% of the Variance Accounted For (VAF). This dimensionality reduction serves as the foundation for our synergy-based models, providing a biologically plausible latent action space that captures essential motor dynamics while discarding high-frequency noise.

### 5.2 High-Level Policy Validation

To isolate strategic planning from motor execution, we validated the Physics-Based High-Level Policy (HLP) in a simplified 6-DOF environment. Testing across three difficulty levels, ranging from fixed ball velocity to stochastic environments with random friction, demonstrated the planner’s robustness (Figure 4). The HLP achieved a 91.5% success rate on the hardest difficulty setting. These results confirm that the primary bottleneck in musculoskeletal table tennis is not strategic placement of the paddle, but the coordination of the high-dimensional muscle system required to achieve those targets.

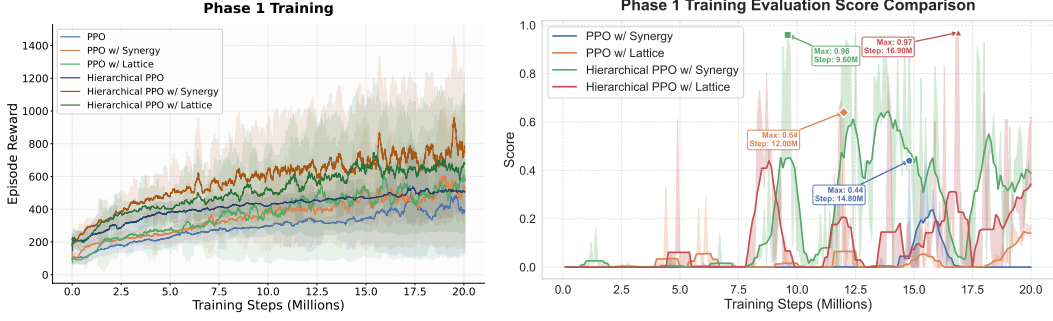


Figure 5: Phase 1 Training Dynamics. **(a)** Mean episode reward over 20M steps; hierarchical models (green/red) dominate flat baselines (blue/orange). **(b)** Evaluation scores showing the rapid convergence of Lattice-enhanced hierarchical models.

### 5.3 Baseline Comparison: Synergy and Latent Exploration

Standard model-free baselines failed to navigate the high-dimensional action space effectively. As illustrated in the Phase 1 Evaluation (Figure 6), the baseline PPO achieved a 0% success rate. However, augmenting the flat action space provided measurable improvements:

- **Muscle Synergy:** Reducing the action space via synergy matrices stabilized the agent, enabling it to achieve a **40% success rate**.
- **Latent Exploration:** Implementing Lattice exploration improved the model more significantly, reaching a **56% success rate**.

While these enhancements outperform the standard baseline, they remain insufficient for the complex table tennis task without hierarchical support.

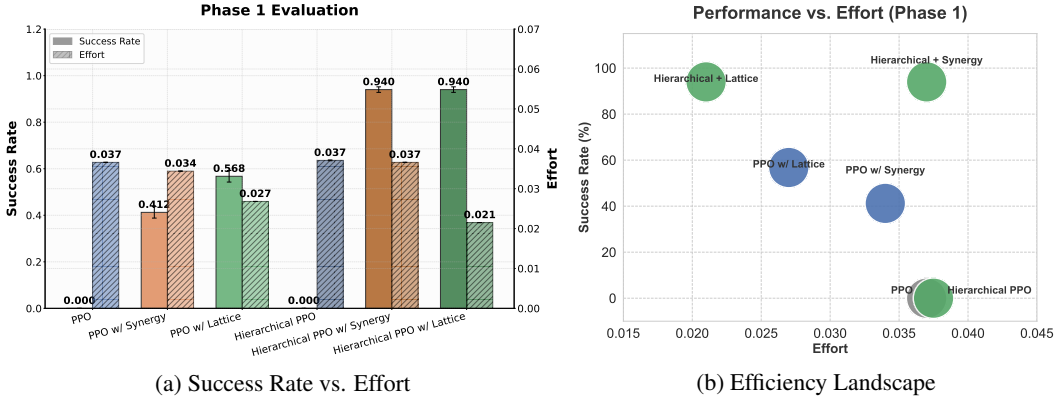


Figure 6: Phase 1 Ablation Results. **(a)** While both hierarchical models achieve near-optimal success (solid bars), the Lattice model uses significantly lower control effort (hatched bars). **(b)** The Pareto frontier view highlights this advantage: the Hierarchical+Lattice model (top-left green bubble) provides the "Best Balance," matching the Synergy model's performance while reducing control effort by  $\sim 43\%$ .

### 5.4 Hierarchical Improvement and Near-Optimal Results

The introduction of a hierarchical architecture reduces the exploration space by allowing the low-level agent to focus entirely on neuromuscular coordination. The Hierarchical PPO architecture consistently outperformed all standard baselines, achieving a **near-optimal success rate of  $\sim 94\%$** . In contrast, non-hierarchical variants remained stalled between 40% and 60%. Critically, hierarchical architectures without any optimization of the action or exploration space (e.g., standard Hierarchical PPO without Synergy or Lattice) failed to achieve successful results, highlighting the importance of combining hierarchical architecture and structured action spaces.



## 5.5 Energy Efficiency, Robustness, and Limitations

While both hierarchical methods reached similar performance peaks, the Lattice approach emerged as the superior solution regarding efficiency. In Phase 1, the **Lattice approach matched the top-tier success rate while requiring significantly less control effort** (0.021 compared to 0.037 for the Synergy model).

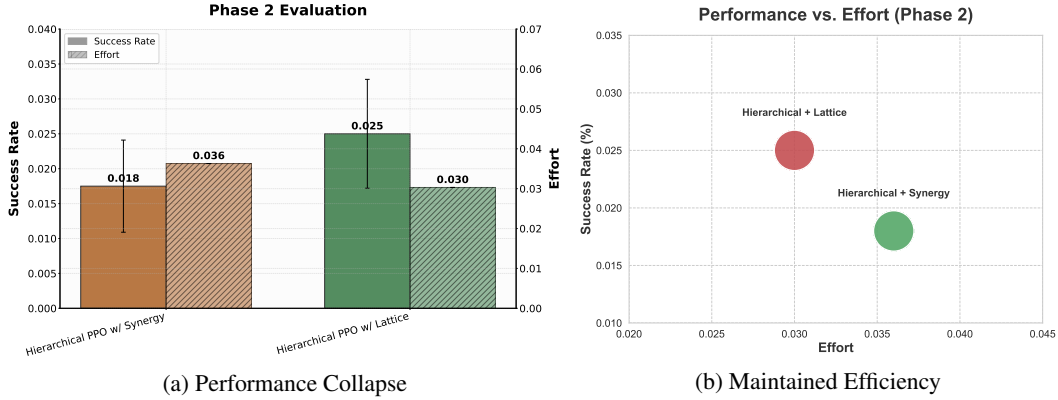


Figure 7: Phase 2 Robustness Results. **(a)** Success rates for both top-performing models drop to the 2–3% range under domain randomization. **(b)** Despite the performance drop, the efficiency characteristic remains consistent: the Lattice model (red bubble) continues to operate at a lower control effort than the Synergy model (green bubble).

However, Phase 2 evaluation (domain randomization of velocity, friction, and mass) revealed existing limitations. As shown in Figure 7, performance for both top-performing models dropped to the 2.5–3% range. Qualitative analysis suggests this collapse is due to:

- **Timing Mismatches:** Difficulty adapting stroke speed to randomized ball velocities.
- **Abrupt Dynamics:** Randomized physical parameters causing unstable movements that destabilize the paddle posture.

Despite the performance drop, the Lattice model maintained its efficiency characteristic, continuing to operate with lower control effort than the Synergy-based counterpart.

## 6 Conclusion and Future Work

This work introduces MyoPong, a hierarchical reinforcement learning framework that effectively masters high-dimensional musculoskeletal control for table tennis. Our results demonstrate that while standard PPO baselines failed completely, integrating structured priors on the action and exploration space enabled the agent to achieve a 94% success rate. Furthermore, our ablation study reveals a key distinction between action space priors and learnable exploration. Although both Muscle Synergy and Lattice methods achieved high task success, the Lattice method proved superior in energy efficiency, offering the best balance of performance and minimal energy cost. This finding suggests that learnable, structured exploration can outperform fixed synergies in complex motor control tasks.

**Limitations and Future Work.** Our evaluation highlighted a significant performance drop under domain randomization, primarily due to timing mismatches where the agent’s stroke speed failed to synchronize with variable ball velocities. Future work may address this by integrating recurrent neural networks (e.g., LSTMs) to better track ball trajectories over time. Additionally, we aim to replace the manually designed High-Level Policy (HLP) with an iterative, LLM-based self-improvement loop to automate strategy generation.

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